

CMSE 201 – Homework 5 Rubric

The total number of points possible in this assignment is: 40

Part 1 (22 points)

- Loading and inspecting the data (3 points):
 1. Import necessary modules – 0.5 point
 2. Read in the data using Pandas – 0.5 point
 3. Show the first 10 lines of the data – 0.5 point
 4. Use describe function to show summary statistics – 0.5 point
 5. Show the column names – 0.5 point
 6. Show the number of rows in the data – 0.5 point

- Determining the maximum expression values and which genes (4 points):
 1. Using mask and max() function to find maximum expression values in CD4 Memory Primary cells – 2 points
 2. Using mask and max() function to show the expression level of that gene in CD4 Naïve Primary cells – 2 points

- Determining the maximum difference of gene expression across two tissues and which genes (4 points):
 1. Using np.max() function to find the maximum difference of gene expression – 2 points
 2. Using mask to identify the specific gene – 2 points

- Finding median values in each tissue (4 points):
 1. Using median() function to print out the median expression values – 2 points
 2. Using mask to find median values for subsets of genes – 2 points

- Visualizing tissue-specific gene expressions using boxplot (3 points):
 1. Creating one additional Boolean column using mask – 1 point
 2. Boxplot of expression values for different groups of genes based on the newly created Boolean column – 1 point
 3. Correct annotations including title, axis labels, y-axis range and one seaborn plot style – 0.5 point
 4. Correct explanation of the meaning of 75% quantile – 0.5 point

- Looking for correlation of gene expression between normal and cancer cells (4 points):
 1. Using combined masks to create a new data frame, including only genes whose expression are non-zero in both normal and cancer liver cells – 1 point
 2. Seaborn jointplot on the new subset data frame – 1 point
 3. Seaborn jointplot on np.log2() transformed data – 1 point

4. Answers to the 2 questions on interpreting the plots and compare the observations: as long as the student give an effort to explain the correlations and the necessity of doing log2 transformation – 1 point

Part 2 (18 points)

- Modeling gene expression in cancer cells as a function of their values from normal cells using curve_fit (9 points):
 1. Define a test_func() – 2 points
 2. Use curve_fit to find the parameters – 2 points
 3. Use test_func() to generate predicted values – 2 points
 4. Scatter plot of the log2 transformed data – 1 point
 5. Plot a line for predicted values – 1 point
 6. Answer to the question about appropriateness of linear model for the observed data (any reasonable interpretations are good) – 1 point

- Modeling gene expression in cancer cells as a function of their values from normal cells using polyfit (9 points):
 1. Use polyfit() to find polynomial parameters with a specified degree – 3 points
 2. Use poly1d() to create the polynomial function – 3 points
 3. Scatter plot of the log2 transformed data – 1 point
 4. Plot a line of predicted values using the polynomial function – 1 point
 5. Answers to the model selection (any reasonable interpretations are good) - 1 point